

Simulazione Molecolare di Materiali

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Lab 1 - WORKSTATION SET-UP AND FIRST FORTRAN PROGRAM

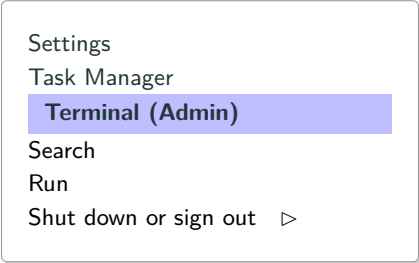
Working Environment: Windows Subsystem for Linux (WSL)

In this course we will work inside a real Linux environment (**Ubuntu**) running directly on your Windows PC, using the **Windows Subsystem for Linux (WSL)**.

- ✎ no dual-boot, no virtual machine: Ubuntu runs alongside Windows.
- ✎ requires Windows 10 (version 2004+) or Windows 11.

Installing WSL and Ubuntu (1/2)

- 🖱 right-click on the Windows **Start** menu button.
- 🖱 select **Terminal (Admin)** (or *Windows PowerShell (Admin)*) from the menu.



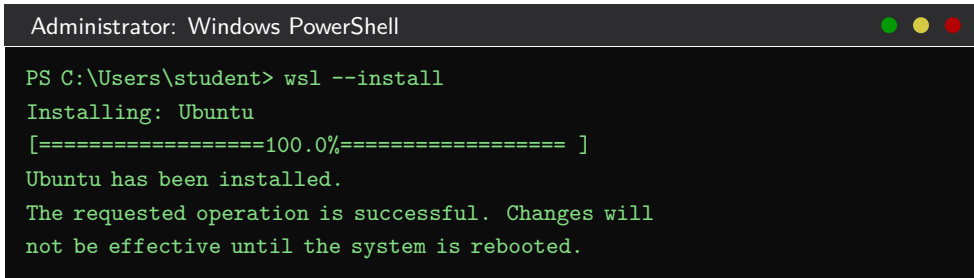
A screenshot of the Windows Start menu search results. The menu is displayed as a list of options: Settings, Task Manager, Terminal (Admin), Search, Run, and Shut down or sign out. The 'Terminal (Admin)' option is highlighted with a blue background. A right-pointing triangle is visible next to the 'Shut down or sign out' option.

Settings
Task Manager
Terminal (Admin)
Search
Run
Shut down or sign out ▷

Installing WSL and Ubuntu (2/2)

In the administrator terminal that opens, run the command:

```
wsl --install
```

A screenshot of a Windows PowerShell terminal window titled "Administrator: Windows PowerShell". The terminal shows the command "wsl --install" being executed. The output indicates that Ubuntu is being installed, shows a progress bar at 100.0%, and confirms that Ubuntu has been installed successfully. It also states that the operation is successful and that changes will be effective after a system reboot.

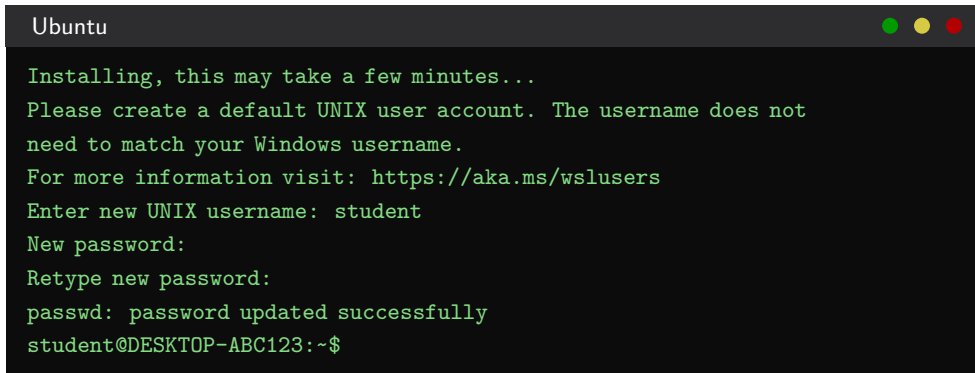
```
Administrator: Windows PowerShell

PS C:\Users\student> wsl --install
Installing: Ubuntu
[=====100.0%===== ]
Ubuntu has been installed.
The requested operation is successful. Changes will
not be effective until the system is rebooted.
```

- 📌 this installs **WSL2** together with the default **Ubuntu** distribution.
- 📌 restart the computer if you are asked to.

First Launch of Ubuntu

- 🖊 after the restart, open **Ubuntu** from the Windows Start menu.
- 🖊 wait for the installation to complete (first launch only).
- 🖊 create a UNIX username and password when prompted (independent from your Windows account – do not forget them!).



```
Ubuntu

Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not
need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: student
New password:
Retype new password:
passwd: password updated successfully
student@DESKTOP-ABC123:~$
```

Updating the System

Before installing anything, always update the package lists and the installed packages:

```
sudo apt update
```

```
sudo apt upgrade -y
```

Installing gfortran

The GNU Fortran compiler is available directly from the Ubuntu repositories:

```
sudo apt install -y gfortran
```

Check the installation with:

```
gfortran --version
```

Installing GROMACS

GROMACS is also packaged for Ubuntu:

```
sudo apt install -y gromacs
```

Check the installation with:

```
gmx --version
```

Installing *xmgrace*

xmgrace is provided by the *grace* package:

```
sudo apt install -y grace
```

Launch it with:

```
xmgrace
```

Installing *gnuplot*

gnuplot is available directly from the Ubuntu repositories:

```
sudo apt install -y gnuplot
```

Launch it with:

```
gnuplot
```

Installing the GitHub CLI

We will use **Git** and the **GitHub CLI** (gh) to download the course material. Both are installed directly from the Ubuntu repositories:

```
sudo apt install -y git gh
```

Check the installation with:

```
gh --version
```

Installing VMD

VMD is **not** available in the Ubuntu repositories: it must be downloaded (free registration required) from the official website:

`https://www.ks.uiuc.edu/Research/vmd/`

Once you have downloaded the Linux binary archive, install it from a terminal:

```
tar xzf vmd-*.tar.gz
cd vmd-*
./configure
cd src
sudo make install
```

- ✎ launch VMD from the terminal simply with: `vmd`
- ✎ to display graphical windows (*gnuplot*, *VMD*, *xmgrace*, ...) launched from WSL, make sure you are using a recent Windows 10/11 build: WSLg (the built-in graphical support) is enabled automatically and requires no extra configuration.

macOS Users - SOFTWARE INSTALLATION VIA HOMEBREW

If you use **macOS**, you do not need WSL: your Mac already has a Unix terminal (*zsh*) built in.

- 📌 open the **Terminal** app (Applications → Utilities → Terminal, or search it with Spotlight).
- 📌 we will use **Homebrew**, the most popular package manager for macOS, to install everything from the terminal.

Installing Homebrew

From the Terminal, run the official installation command:

```
/bin/bash -c "$(curl -fsSL  
↪ https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

- 📌 the Xcode Command Line Tools will be installed automatically if missing.
- 📌 at the end, follow the on-screen instructions to add Homebrew to your PATH.

Installing gfortran (via GCC)

The GNU Fortran compiler is distributed together with GCC:

```
brew install gcc
```

🔗 Homebrew installs a **versioned** binary, e.g. `gfortran-15` (the exact number depends on the current GCC release).

🔗 find the exact name with:

```
ls $(brew --prefix)/bin/gfortran-*
```

Installing gfortran: Optional Alias

To use the plain `gfortran` command (matching the rest of this course), create a symlink (replace 15 with your installed version):

```
ln -s $(brew --prefix)/bin/gfortran-15 $(brew --prefix)/bin/gfortran
```

Check the installation with:

```
gfortran --version
```

Installing GROMACS

GROMACS provides a native Homebrew formula, with pre-built binaries for Apple Silicon:

```
brew install gromacs
```

Check the installation with:

```
gmx --version
```

Installing *xmgrace*

xmgrace is also available via Homebrew, but being an X11 application it requires **XQuartz**:

- ✎ install XQuartz from: <https://www.xquartz.org>
- ✎ log out and log back in (or restart) after installing XQuartz.
- ✎ install *grace*:

```
brew install grace
```

Launch it with:

```
xmgrace
```

Installing *gnuplot*

gnuplot is available via Homebrew:

```
brew install gnuplot
```

Launch it with:

```
gnuplot
```

Installing the GitHub CLI

We will use **Git** and the **GitHub CLI** (gh) to download the course material:

```
brew install git gh
```

Check the installation with:

```
gh --version
```

Installing VMD

- 📌 VMD is **not** available via Homebrew: download it (free registration required) from the official website:

`https://www.ks.uiuc.edu/Research/vmd/`

- 📌 choose the **macOS** build for your Mac (Apple Silicon or Intel).
- 📌 open the downloaded package and follow the installer to copy **VMD.app** into your **Applications** folder.
- 📌 launch it from Launchpad/Applications, or from the terminal with: `open -a VMD`

Getting the Course Material - CLONING THE EXERCISES REPOSITORY

GitHub Authentication

Before cloning, you need to log in once with the GitHub CLI (works the same on WSL/Ubuntu and macOS):

```
gh auth login
```

- 📌 choose **GitHub.com**, then **HTTPS**.
- 📌 choose **Login with a web browser**.
- 📌 copy the one-time code shown in the terminal and paste it into the browser page that opens.

Cloning the Course Repository

The exercises for this course are hosted on GitHub. From your home directory, run:

```
gh repo clone antdenicola/corso-simulazione-molecolare-materiali
```

📌 this creates a local folder `corso-simulazione-molecolare-materiali` containing all the exercises.

📌 move into it with:

```
cd corso-simulazione-molecolare-materiali
```

LINUX – Essential Commands

- 📌 `ls`: lists the files in a directory. If no parameter is given, it lists only the names of all files in the current directory.
- 📌 `pwd`: *print working directory*.
- 📌 `cd`: *change directory*; used to change the current working directory.
- 📌 `mkdir`: used to create a new directory.
- 📌 `cp`: used to copy a file. `cp [file to copy] [destination]`
- 📌 `mv`: used to move a file. `mv [file to move] [destination]` *!!! Warning: if the file being moved has the same name as the destination one (e.g. in a different directory), it will be overwritten.*
- 📌 `rm`: **irreversibly** deletes a file *!!!* `rm [file to delete]`

-  `i` insert text before the cursor
-  `dd` delete the current line
-  `:q!` quit the editing session without saving
-  `:wq!` quit the editing session saving changes

Fortran Language (FORmulaTRANslator)

A programming language allows the operations that the computer must perform to be translated into a program.

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Three main phases

1. Writing the Program
2. Compilation
3. Execution

Fortran Language (FORmulaTRANslator)

A programming language allows the operations that the computer must perform to be translated into a program.

Three main phases

1. Writing the Program
2. Compilation
3. Execution
 - In the writing phase the commands are given. You need to know the rules (spelling, syntax) of the language you are using.
 - In the compilation phase the program text (called the *source*) is translated by the compiler into a language the computer can understand (a binary file).
 - The program is executed by running the binary file.

First Exercise: The Fortran Language

An easy example: the sum of two numbers.

Note

A general rule of Fortran is that the program text must start at the seventh column (a legacy of punched cards).

```
Program Sum
```

```
write(6,*) 'enter the first number'
```

```
read(5,*) a
```

```
write(6,*) 'enter the second number'
```

```
read(5,*) b
```

```
s = a + b
```

```
write(6,*) 'the sum is'
```

```
write(6,*) s
```

```
stop
```

```
end
```

First Exercise: The Fortran Language

Input/Output Units

- the number **5** denotes an **input** unit, in this case the keyboard.
- the number **6** denotes an **output** unit, in this case the screen.

First Exercise: The Fortran Language

Compilation and execution:

-  edit the file: `vim sum.f`
-  enter the program text and save
-  compile: `gfortran -o sum sum.f`
-  run: `./sum`
-  enter the first number and press Enter
-  enter the second number and press Enter

First Exercise: The Fortran Language

The program can easily be modified to perform other operations, or a combination of these:

-  product `*`
-  division `/`
-  square root `sqrt(a)`
-  exponentiation `a**2.0` (a squared)
-  sine `sin(a)` (radians)
-  cosine `cos(a)`
-  natural logarithm `ln(a)`